

Learning the Inverse Kinematics of Tendon-driven Soft Manipulators with K-nearest Neighbors Regression and Gaussian Mixture Regression

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Abstract—Due to the urgent need for Minimally Invasive Surgeries (MIS), all kinds of surgical robots have been developed and investigated intensively in last decades, which can both release the fatigues of surgeons and speed up the process of wound healing. Tendon-Driven Serpentine Manipulator (TSM) maybe among the most widely adopted and promising ones to turn robot assisted MIS into reality. But due to the high nonlinearities and model uncertainties in the TSM system, it is extremely difficult to precisely control the robot. In this paper, we develop and investigate two approaches from machine learning domain, Gaussian Mixture Regression (GMR) and K-Nearest-Neighbors Regression (KNNR), to learn the Inverse Kinematic (IK) model of our TSM robot. Then we compare the performance of GMR and KNNR with that of an IK model derived in previous literatures. Experimental results conducted on a real world TSM robot performing trajectory tracking tasks validate the superior performance of the proposed methods over traditional analytical IK models.

Keywords—Gaussian mixture regression; human robot interaction; inverse kinematic model; K-nearest neighbors regression; machine learning; surgical manipulators

I. INTRODUCTION

Due to the rapid and intense development of robot assisted Minimally Invasive Surgeries (MIS), a lot of different kinds of flexible and continuum robots for surgical purposes have been designed and investigated in the last decades. And Tendon-driven Serpentine Manipulator (TSM) [1], Concentric Tube Robot (CTR) [2] and pneumatically driven soft tentacles [3] may be among the most popular ones. However, due to the high nonlinearities inherent in these flexible robots, precise kinematic or dynamic control becomes extremely difficult. There are several reasons resulting in this dilemma. Firstly, it is usually analytically inaccessible to derive an accurate kinematic model for the flexible robots which makes model-based control strategy impossible. Secondly, these flexible robots usually incorporate elastic beam elements to enable continuum bending, and introducing nonlinearities to the system. More specifically, for tendon driven manipulator which is the focus in this paper, cable tensioning and slack cause friction and backlash in the system. Therefore, these systems usually exhibit high nonlinearities and conventional modeling

approach based on the assumption of constant curvature is insufficient.

Precise motion control is of high importance especially in surgical applications like targeting diseased location or dissecting diseased tissue. In order to improve motion control accuracy for flexible surgical manipulators, researchers have designed feed-forward controllers to compensate for part of the nonlinear effects by modeling friction and backlash [4], [5]. With miniaturized position sensors and vision sensors playing more and more important roles in surgical robotics, feedback control were also investigated in last decade. PID controller is one of the earliest attempts [6], [7], which greatly enhances the control accuracy comparing to open loop. The disadvantages are in two folds: the control loop performs poorly when the manipulator has to change the bending direction; actuation commands and control action decomposition depends on the inaccurate kinematic model. Intelligent control schemes such as fuzzy logic were later proposed [8], [9], which relies on a reliable inverse dynamic model. The design of multiple membership functions can be troublesome.

As analytical solution is usually unattainable and numerical solution requires exhaustive iterative process, data-driven approach has proven to be a more efficient and reasonable alternative to learn the inverse kinematic model of TSM robot. Machine learning algorithms are efficient at approximating nonlinear functions provided with proper training dataset and well selected input features. Feed-forward Neural Network (FNN) has been applied to learn the nonlinear inverse kinematic model of CTR [10] and soft tendon driven manipulators [11], [12]. Another regression method based on FNN called ELM is applied to train the inverse equilibrium model of a soft arm in a faster manner based on prior knowledge [13]. In this work, we propose to use two popular regression methods, namely Gaussian Mixture Regression (GMR) [14] and K-Nearest Neighbors Regression (KNNR) [15] to learn the highly nonlinear inverse kinematic model of our TSM robot. The performance of the algorithms were evaluated by real world trajectory tracking experiments.

The rest of the paper is structured as follows: Section II introduces the forward and inverse kinematic model of the tendon-driven serpentine manipulator. Section III describes Gaussian Mixture Regression and K-Nearest Neighbors Regression in detail. Section IV evaluates the performance of

the learned models by trajectory tracking experiments on our TSM robot and at the same time compares the results with that of an inverse kinematic model derived in previous literatures [16]. Section V discusses the experimental results and Section VI draws conclusions of this work and introduces future work.

II. FORWARD AND INVERSE KINEMATIC MODEL OF THE TENDON-DRIVEN SERPENTINE ROBOT

In recent years, the flexible TSM has emerged as one of the most promising platform technologies for Minimally Invasive Surgeries (MIS). Comparing to its rigid counterparts, the flexible manipulator shows greater advantages in navigating through confined and torturous human lumen. Therefore, an in-house master-slave TSM robotic platform utilizing such mechanism has been developed for transoral and upper GI tract surgical procedures.

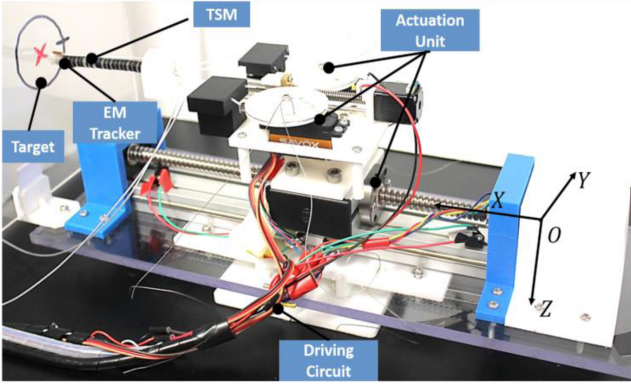


Figure 1. Experimental setup of the 3 DOF TSM robot.

The schematic of our TSM is shown in Fig. 1. The bending section is composed of a series of identical rigid vertebrae with an outer diameter of 7.5mm, with the adjacent two forming a ball joint. An elastic tube is inserted into the central cavity to regulate the bending curvature. The flexible arm is articulated by four orthogonally arranged tendons. The TSM is placed on a moving stage for translational motion. In ideal cases, namely no nonlinear uncertainties, the kinematic model of the flexible TSM can be derived under constant curvature assumptions [17]. The modeling procedure requires mapping among three spaces, namely the actuation space characterized by tendon length variations l_1 , l_2 and base translation Zb ; the configuration space parametrized by θ and φ ; and the task space denoted by the position vector (x, y, z) of the end-effector. Specifically, in trajectory tasks, we are interested in the instantaneous kinematics and associated Jacobian matrix of the system, which is given as follows:

$$\dot{l} = J_1 \dot{q} \quad (1)$$

$$\dot{P} = J_2 \dot{q} \quad (2)$$

where $\dot{q} = [\dot{\theta} \ \dot{\varphi}]^T$, $\dot{l} = [\dot{l}_1 \ \dot{l}_2 \ \dot{Z}b]^T$, and $\dot{P} = [\dot{x} \ \dot{y} \ \dot{z}]^T$, J_1 relates the configuration space with the actuation space,

while J_2 relates the task space with the configuration space and they can be explicitly written as:

$$J_1 = \begin{pmatrix} Ndc_{\bar{\theta}}c_{\varphi}/2 & -Nds_{\bar{\theta}}s_{\varphi} \\ Ndc_{\bar{\theta}}s_{\varphi}/2 & Nds_{\bar{\theta}}c_{\varphi} \\ -Nc_{N\bar{\theta}}c_{(N+1)\bar{\theta}}/2s_{\bar{\theta}} & \\ -c_{\bar{\theta}}s_{N\bar{\theta}}c_{(N+1)\bar{\theta}}/2s_{\bar{\theta}}^2 & 0 \\ +(N+1)s_{N\bar{\theta}}s_{(N+1)\bar{\theta}}/2s_{\bar{\theta}} \end{pmatrix} \quad (3)$$

$$J_2 = \frac{l}{2s_{\bar{\theta}}} \begin{pmatrix} c_{\varphi}[Nc_{N\bar{\theta}}s_{(N+1)\bar{\theta}} \\ -c_{\bar{\theta}}s_{N\bar{\theta}}s_{(N+1)\bar{\theta}}/s_{\bar{\theta}} & -2hs_{N\bar{\theta}}s_{\varphi}s_{(N+1)\bar{\theta}} \\ +(N+1)s_{N\bar{\theta}}c_{(N+1)\bar{\theta}}] \\ s_{\varphi}[Nc_{N\bar{\theta}}s_{(N+1)\bar{\theta}} \\ -c_{\bar{\theta}}s_{N\bar{\theta}}s_{(N+1)\bar{\theta}}/s_{\bar{\theta}} & 2hs_{N\bar{\theta}}c_{\varphi}s_{(N+1)\bar{\theta}} \\ +(N+1)s_{N\bar{\theta}}c_{(N+1)\bar{\theta}}] \\ Nc_{N\bar{\theta}}c_{(N+1)\bar{\theta}} \\ -c_{\bar{\theta}}s_{N\bar{\theta}}c_{(N+1)\bar{\theta}}/s_{\bar{\theta}} & 0 \\ +(N+1)s_{N\bar{\theta}}s_{(N+1)\bar{\theta}} \end{pmatrix} \quad (4)$$

Let $J = J_2 J_1^* \in \mathbb{R}^{3 \times 3}$, where J_1^* is the pseudo-inverse of J_1 , be the system Jacobian matrix relating the robot task space with the actuation space. J is non-singular for $\forall \bar{\theta} \neq 0$.

III. METHODS

A. Data Acquisition and Preprocessing

With the inverse kinematic model described in Section II, the TSM robot is controlled to track circles with different diameters, but due to the high nonlinearities and model uncertainties unconsidered in the IK model, the accuracy of outcome trajectories is very poor (far from reference circles), see Fig. 3(c). In order to improve the tracking performance of TSM robot, we implement GMR and KNNR in this section. Before passing the acquired data to the learning algorithms, preprocessing of the data is conducted. The Savitzky-Golay filter is applied to smooth the data and delete outliers, dynamic time warping is then used to align the data in the time domain. After preprocessing, the data is transferred to GMR and KNNR to learn an accurate inverse kinematic model of the TSM robot.

B. Gaussian Mixture Models and Gaussian Mixture Regression

The training data set $\xi = \{\xi_i\}_{i=1}^N$ consists of N pairs of observations of the motor movements (left and right DC servo motors) and end-effector states from human demonstrations. Within each data pair, $\xi_m \in \mathbb{R}^2$ denotes the movement of the two servo motors, $\xi_e \in \mathbb{R}^2$ represents the state of the end-effector and ξ_i can be rewritten as $\xi_i = (\xi_e, \xi_m)_i \in \mathbb{R}^4$.

To encode the observed data, GMM is expressed by the following probability density function:

$$p(\xi_i) = \sum_{k=1}^K p(k) N(\xi_i | \mu_k, \Sigma_k) \quad (5)$$

$$N(\xi_i | \mu_k, \Sigma_k) = \frac{1}{\sqrt{(2\pi)^3 |\Sigma_k|}} \exp \left[-\frac{(\xi_i - \mu_k)^T \Sigma_k^{-1} (\xi_i - \mu_k)}{2} \right]$$

$$\sum_{k=1}^K p(k) = 1$$

where K is the number of Gaussian components in GMM, $p(k)$ is the prior probability of the k^{th} component and $N(\xi_i | \mu_k, \Sigma_k)$ is a Gaussian distribution, μ_k is the mean vector, and Σ_k is the covariance matrix. Given the training data and formulas of GMM, the Expectation Maximization (EM) algorithm [18] is applied to estimate the parameters of GMM, and k-means [19] is applied to initialize the EM algorithm.

GMR is used to extract a generalized motor trajectory from the GMM. The output motor commands ξ_m can be computed from the joint probability distribution of GMM and take the expected values. Robot end-effector state ξ_e is used to query for the estimation of corresponding motor commands. Since $\xi_i = (\xi_e, \xi_m)_i$, μ_k and Σ_k can also be written by:

$$\mu_k = (\mu_{ek}, \mu_{mk}) \quad (6)$$

$$\Sigma_k = \begin{bmatrix} \Sigma_{eek} & \Sigma_{emk} \\ \Sigma_{mek} & \Sigma_{mmk} \end{bmatrix} \quad (7)$$

Given each component k and end-effector state ξ_e , the conditional probability distribution of ξ_m can be obtained:

$$p(\xi_m | \xi_e, k) = N(\xi_m | \tilde{\xi}_{mk}, \tilde{\Sigma}_{mmk}) \quad (8)$$

$$\tilde{\xi}_{mk} = \mu_{mk} + \Sigma_{mek} \Sigma_{eek}^{-1} (\xi_e - \mu_{ek}) \quad (9)$$

$$\tilde{\Sigma}_{mmk} = \Sigma_{mmk} - \Sigma_{mek} \Sigma_{eek}^{-1} \Sigma_{emk} \quad (10)$$

The conditional probability of ξ_m with respect to ξ_e can be defined by:

$$p(\xi_m | \xi_e) = \sum_{k=1}^K \beta_k N(\xi_m | \tilde{\xi}_{mk}, \tilde{\Sigma}_{mmk}) \quad (11)$$

$$\beta_k = \frac{p(k)p(\xi_e | k)}{\sum_{k=1}^K p(k)p(\xi_e | k)} = \frac{p(k)N(\xi_e | \mu_{ek}, \Sigma_{eek})}{\sum_{k=1}^K \pi_k N(\xi_e | \mu_{ek}, \Sigma_{eek})} \quad (12)$$

According to the equations above, ξ_m at end-effector state ξ_e can be estimated by:

$$\tilde{\xi}_m = \sum_{k=1}^K \beta_k \tilde{\xi}_{mk} \quad (13)$$

$$\tilde{\Sigma}_{mm} = \sum_{k=1}^K (\beta_k)^2 \tilde{\Sigma}_{mmk} \quad (14)$$

Therefore, at different end-effector state ξ_e , a smooth generalized form of the motor movement $\tilde{\xi}_m$ can be calculated with this method.

C. K-Nearest Neighbors Regression

In the field of pattern recognition and machine learning, K-Nearest Neighbors Algorithm (KNN) is among the simplest ones, while it is also one of the most efficient methods. Many techniques are developed to enhance the performance of KNN over last decades, such as weighted KNN, condensed KNN, tunable KNN, model based KNN and so on. The goal of KNN in regression problem is to learn a function $f_{KNNR}(\xi_e): \xi_e \rightarrow \xi_m$ from N observed input-output samples $[(\xi_e^1, \xi_m^1), (\xi_e^2, \xi_m^2), \dots, (\xi_e^N, \xi_m^N)]$:

$$f_{KNNR}(\xi_e) = \frac{\sum_{k=1}^K \|\xi_e - \xi_{ek}\|_2 \tilde{f}_{KNNR}(\xi_{ek})}{\sum_{k=1}^K \|\xi_e - \xi_{ek}\|_2} \quad (15)$$

In fact, KNNR is a locally weighted regression method, just like Locally Weighted Projection Regression (LWPR) [20] and Locally Gaussian Process Regression (LGPR) [21], but it also suffers from the same problems LWPR and LGPR have. They all need to store all the training data for future prediction use which costs a lot of memory space.

In the next section, experimental results of GMR and KNNR for a TSM robot prototype tracking a semi-circle are shown and discussed. We also compare the results of GMR and KNNR with those from an analytical IK model derived in previous literatures.

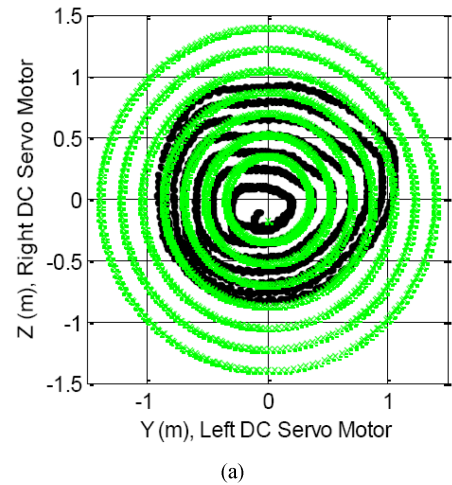
IV. EXPERIMENTAL RESULTS

A. Inverse Kinematics Learning with GMM

Gaussian Mixture Regression is a multivariate nonlinear regression approach which can approximate a continuous function from a multidimensional input ξ_e to a multidimensional output ξ_m . Given the training data and formulas of GMM, the Expectation Maximization (EM) algorithm is applied to estimate the parameters of GMM, and k-means algorithm initializes the EM algorithm.

$$\xi_m = \tilde{f}_{GMR}(\xi_e) = \sum_{k=1}^K H^k(\xi_e)(A^k \xi_e + B^k) \quad (16)$$

where ξ_m is the motor movements, ξ_e is the end-effector state, K is the number of Gaussian distributions in GMM, the others are defined as in Section II. Fig. 2(a) shows the inputs (end-effector states) and outputs (motor movements), Fig. 2(b) illustrates the results of Gaussian Mixture Model, where $K=9$, and Fig. 2(c) demonstrates the output of Gaussian Mixture Regression to generalize motor movements with respect to a given end-effector trajectory, the prediction accuracy for left motor and right motor are 87.3900% and 95.0400%, respectively.



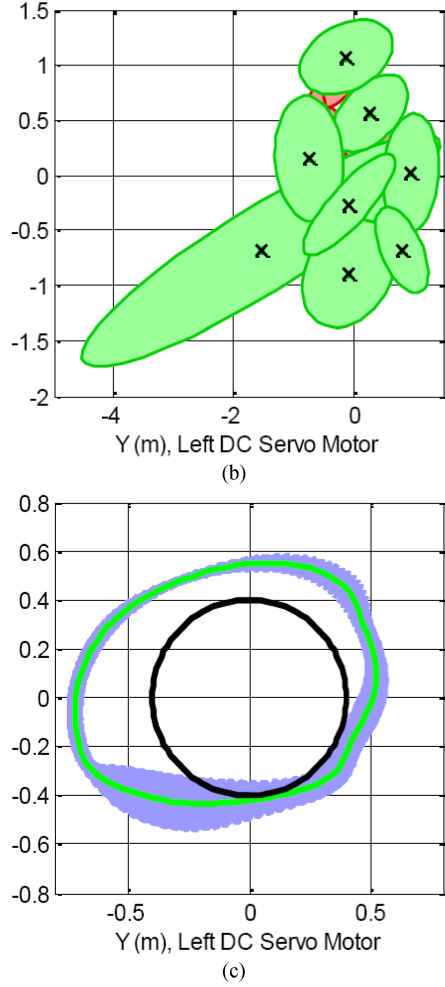


Figure 2. Results of Gaussian Mixture Model and Gaussian Mixture Regression to encode DC servo motor commands and robot end-effector states into a joint probabilistic model. (a) 2D illustration of recorded servo motor commands and robot end-effector states during training period, the black dots denote robot end-effector states while the green ones are the corresponding servo motor commands. (b) Encode servo motor commands and robot end-effector states into a single joint probabilistic model with GMM. (c) The black line denotes a reference circle, and the green line is corresponding servo motor commands generalized from GMR, the blue shadow represents covariance.

B. Inverse Kinematics Learning with KNNR

The K-Nearest-Neighbors Regression (KNNR) is applied in this section to make the predictions for the testing set. KNNR is an instance-based learning algorithm for regression, which do not need pre-training process. KNNR calculates the distance between the testing sample and every sample in the training set based on their features, then it finds the K training samples closest to the testing sample. With these K training samples, KNNR can define the predicted output for the testing sample. The KNNR model can be expressed by:

$$\tilde{f}_{KNNR}(\xi_e) = \frac{\sum_{k=1}^K \|\xi_e - \xi_{ek}\| \tilde{f}_{KNNR}(\xi_{ek})}{\sum_{k=1}^K \|\xi_e - \xi_{ek}\|} \quad (17)$$

Generally speaking, KNNR is able to approximate any function, however, when the dimension of data increases, it can be easily fooled in the curse of dimensionality and dimensionality reduction techniques are often used to solve this problem. For more details, please refer to [16], [17]. The prediction accuracy for the left DC servo motor and the right DC servo motor are 90.4957% and 95.9142%, respectively. And Fig. 3 (b) demonstrates the output trajectory of KNNR to generalize motor movements with respect to a given end-effector trajectory.

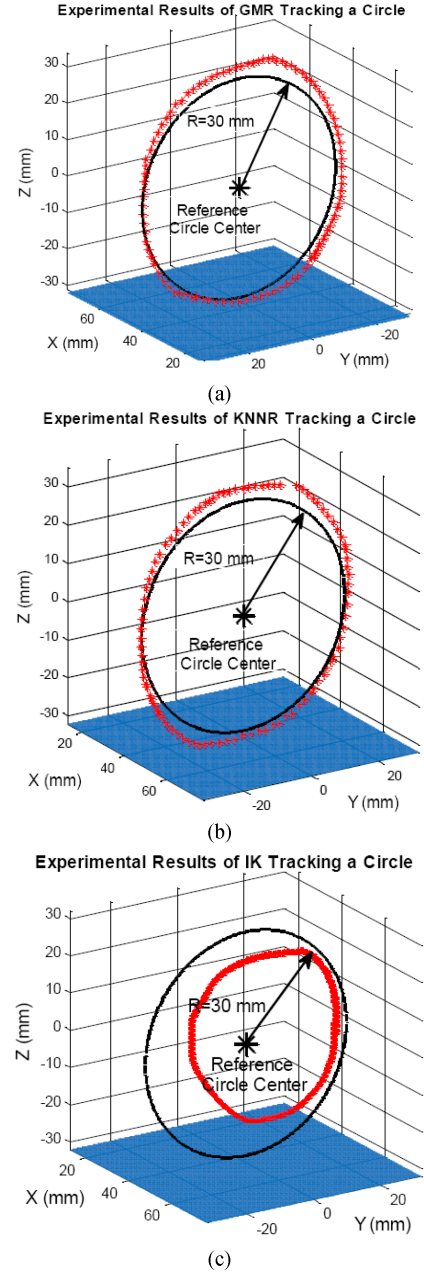


Figure 3. Experimental results of the proposed three approaches on tracking a circle with TSM robot in 2D space. (a) Results of GMR. (b) Results of KNNR. (c) Results of Inverse kinematics method. We can clearly see that KNNR has the highest accuracy on real world robot.

V. DISCUSSION

GMR has the lowest prediction accuracy, 87.39% for the left servo motor and 95.04% for the right servo motor. While KNNR performs better during the training and testing period, the prediction accuracy is about 90.4597% for the left servo motor and 95.9142% for the right servo motor.

For the real world experiment, both algorithms are required to generalize motor movements for the TSM robot to track a circle with the radius 30mm, KNNR outperforms GMR, detailed information can be found in Fig. 3. We also compare the results of GMR and KNNR with that of an IK model derived in previous literatures [16], and KNNR is still demonstrated to be the best and may be the most suitable method to learn the inverse kinematics of TSM robot. The Rooted Mean Square Error (RMSE) of GMR, KNNR and IK are 2.5556mm, 2.1527mm and 8.3359mm, respectively.

VI. CONCLUSION

In this work, two heuristic methods from the statistical machine learning domain, namely Gaussian Mixture Regression (GMR) and K-Nearest-Neighbors Regression (KNNR), have been developed to learn the inverse kinematics of TSM robots. We compare the experimental results of GMR and KNNR with that of an inverse kinematic model derived in previous literatures on a trajectory tracking task, both GMR and KNNR are better than the previously derived IK model in the tracking task. Moreover, as GMR and KNNR are all heuristic approaches, they can be easily transferred to different robotic platforms.

Future work will be focused on two aspects: first, reinforcement learning will be investigated to iteratively update the parameters of GMR and KNNR based on reward function. Second, the proposed methods will be combined with differential dynamical systems to enable motion planning and obstacle avoidance for the TSM robot.

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